HALDIA INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION UNDER MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL)

Paper Code : 001-PCC-CSE404-N

Paper Name : Discrete Mathematics

Time Allotted : 3 Hours *Full Marks : 70*

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Group-A
(Multiple Choice Type Questions)

[15 x 1 = 15]

1. Choose the correct alternatives from the followings:

(I) Let $p(x)$: "$x \in N$ such that $x^2 + 2 < 12$". The truth set of $p(x)$ is CO-3

(a) $\{x: x \in N, x > 5\}$

(b) $\{x: x \in N, x \leq 5\}$

(c) $\{x: x \in N, x > 3\}$

(d) None of these

(II) How many solutions are there to the congruence $15x \equiv 25 \ (mod\ 35)$ CO-3

(a) 3

(b) 1

(c) 0

(d) 5

(III) The root of the binary tree having degree CO-4

(a) 1

(b) 2

(c) 3

(d) 4

(IV) The chromatic number of a complete graph with 7 vertices is CO-2

(a) 5

(b) 6

(c) 7

(d) 0

(V) What is the reminder when the sum $1! + 2! + 3! + \cdots + 100!$ Is divided by 18 CO-3

(a) 12

(b) 6

(c) 9

(d) None of these

(VI) A connected planar determines how many regions it has if it has 5 edges and 3 vertices CO-1

(a) 3

(b) 4

(c) 5

(d) 6

(VII) A loop in G gives which one of the following in its dual CO-2

(a) A pair of parallel edges

(b) Loop

(c) An even vertex

(d) None of these

(VIII) If G is non-planar graph, then the possible number of vertices of G is CO-2

(a) 4

(b) 5

(c) 6

(d) 3

(IX) The number of vertices of a complete binary tree having height 5 is CO-4

(a) 31

(b) 15

(c) 63

(d) None of these

(X) If a finite set has 5 elements, then the number of elements of its power set is CO-1

(a) 5!

(b) 10

(c) 12

(d) 32

(XI) Find the minimum number of people out of 25 must have their birthday in the same month when they are assembled in a room. CO-1

(a) 1

(b) 2

(c) 3

(d) 4

(XII) How many committees of 5, with a given chairperson, can be selected from 12 persons? CO-1

(a) 12

(b) 120

(c) 3960

(d) None of these

(XIII) A group G is commutative iff for all a, b, it satisfies CO-3

(a) ab=ba

(b) $(ab)^{-1} = b^{-1}a^{-1}$

(c) $(ab)^{-1} = a^{-1}b^{-1}$

(d) None of these

(XIV) How many ways can a tree with 5 vertices be colored with at most 4 colors? CO-4

(a) 322

(b) 12

(c) 324

(d) None of these

(XV) Integers less than 12 and relatively prime to 12 are CO-2

(a) 1, 2, 3, 4, 6, 12

(b) 2, 4, 6, 9, 10

(c) 1, 5, 7, 11

(d) 5,7,11

Group-B

(Short Answer Type Questions)

[3 x 5 = 15]

2(I) Obtain the DNF of $\sim\{\sim(p \leftrightarrow q) \wedge r\}$ [5] CO-1

OR

2(II) State and prove Euler's formula for a connected planar graph. [5] CO-1

3(I) What is the rank of the word "GOOGLE" written down in a dictionary? [5] CO-2

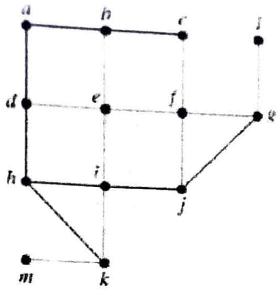
OR

3(II) Find the range set of the function $f(x) = \frac{x}{x^2+1}$. [5] CO-2

4(I) Prove that the following equivalence $(q \wedge \sim (p \wedge q)) \vee (p \wedge (\sim p \vee q)) \equiv q$. ! [5] CO-4

OR

4(II) Find a spanning tree starting from for the following graph using breadth-first search algorithm [5] CO-4



Group-C

(Long Answer Type Questions) [4 x 10 = 40]

5(I) (a) Find the number of words that can be formed from the letters of the word "ELEVEN." (a) How many of them begin and end with E? (b) How many of them have the three E's together? (c) How many begin with E and end with N? [5] CO-1

(b) Find the total number of integers between 1 and 1000 which are neither perfect squares nor perfect cubes. [5]

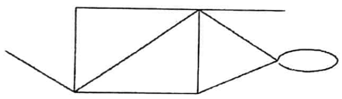
OR

5(II) (a) Show that $(q \rightarrow (p \wedge {\sim}p)) \rightarrow (r \rightarrow (p \wedge {\sim}p)) \equiv (r \rightarrow q)$. [5] CO-1

(b) Using the truth table show that $p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$. [5]

6(I) (a) State and prove the generalized Euler theorem for planar graphs. [5] CO-2

(b) Draw the dual of the following graph [5]



OR

6(II) Show that the set of rational numbers other than one, $Q' = Q - \{1\}$ forms a group under the binary operation $*$ defined by $a * b = a + b - ab, \forall\, a, b \in Q'$. [10] CO-2

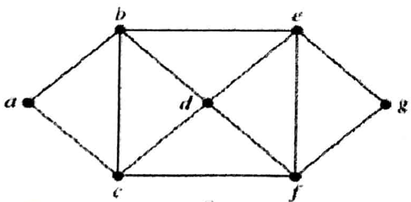
7(I) Applicants a_1, a_2, a_3, a_4 apply for five posts p_1, p_2, p_3, p_4 and p_5. The applications are done as follows: $a_1 \rightarrow \{p_1, p_2\}$, $a_2 \rightarrow \{p_1, p_3, p_5\}$, $a_3 \rightarrow \{p_1, p_2, p_3, p_5\}$ and $a_4 \rightarrow \{p_3, p_4\}$. (i) Draw a bipartite graph of the relationship between applicants and posts (ii) Is there any perfect matching of the set of applicants into the set of post? If yes use Hall's marriage theorem to show that every applicant can be offered a single post. [10] CO-3

OR

7(II) (a) Prove that the intersection of two subgroups is always a subgroup, while the union of two subgroups is not necessarily a subgroup. [5] CO-3

(b) Find the generators and order of each element of the group $G = \{a, a^2, a^3, \ldots, a^8 = e\}$. [5]

8(I) (a) Find three different independent set of vertices with different sizes, and also find the chromatic number of the graph given below [5] CO-4



(b) If the number of edges of a simple planar graph G is four greater than the number of vertices, then find the number of regions determined by G. [5]

OR

8(II) (a) Find the inverse of the permutation: (a) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$. [5] CO-4

(b) Let $G = (Z, +), G' = (Z, +)$ and $f: G \rightarrow G'$ defined by $f(x) = |x|$. Verify whether f is a homomorphism. If so, determine $Ker\ f$. [5]

*** END OF PAPER ***